

AudioSense: Leveraging Current-to-Acoustic Channel to Detect Appliances at Single-Point

Yijie Li[#], Xiatong Tong[#], Qianfei Ren[#], Qingyang Li[#], Lanqing Yang[#],
Yi-Chao Chen[#], Guangtao Xue[#], Xiaoyu Ji^{*}, Jiadi Yu[#]

Shanghai Jiao Tong University[#]

Zhejiang University^{*}



上海交通大学

SHANGHAI JIAO TONG UNIVERSITY



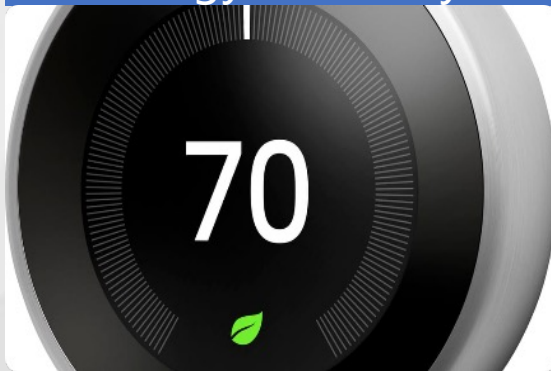
浙江大学

Zhejiang University

Motivation

Electrical appliances monitoring is widely used in the smart home ecosystem.

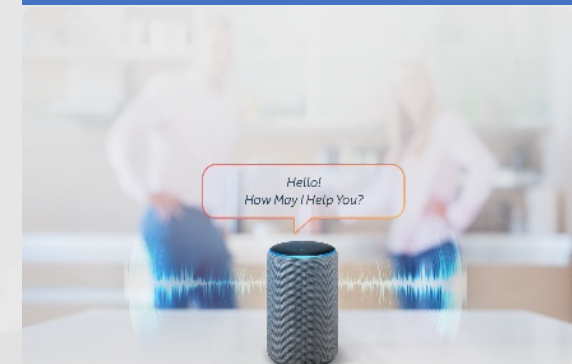
Energy Efficiency



Activity Recognition



Human Interaction



Fire Prevention

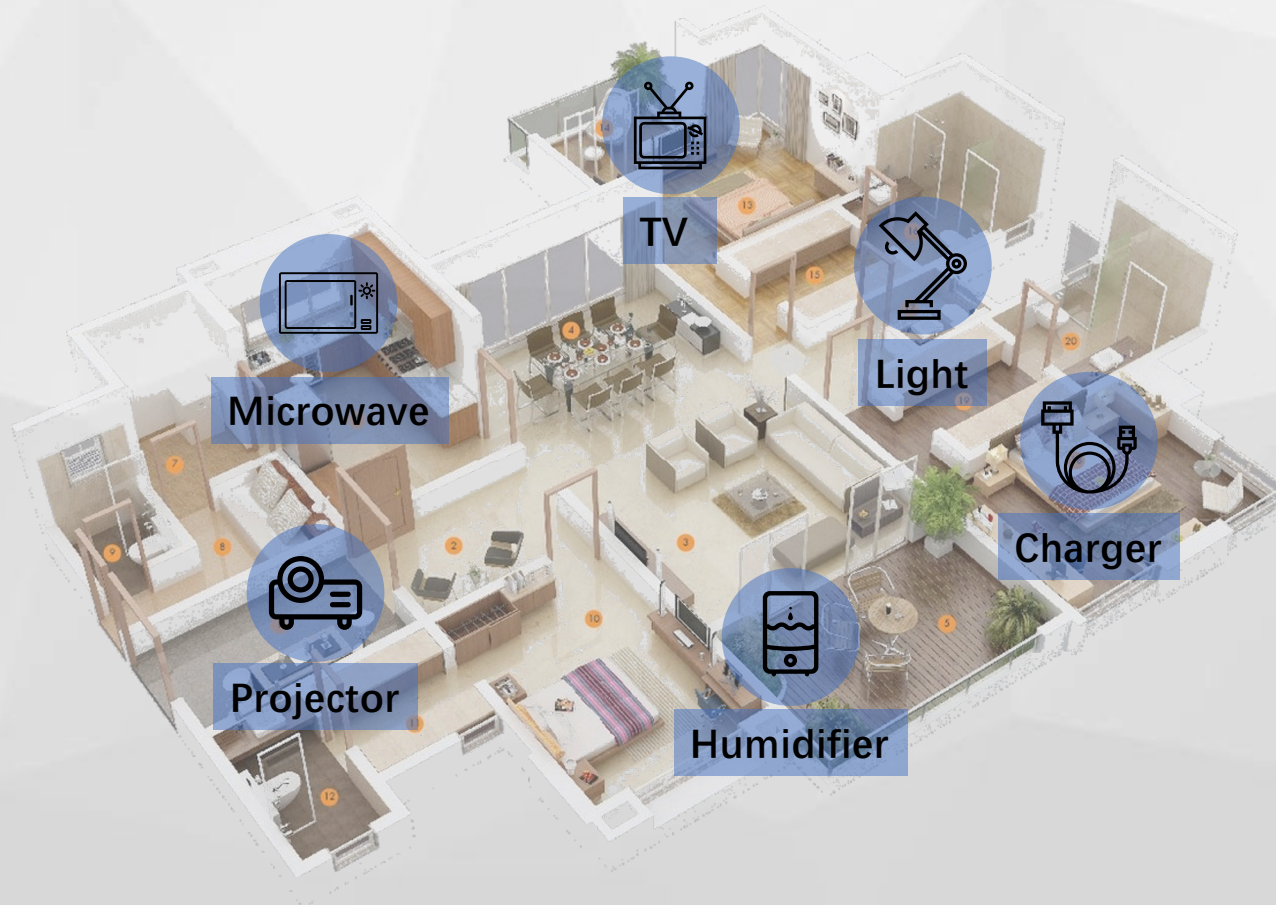


Anomaly Detection

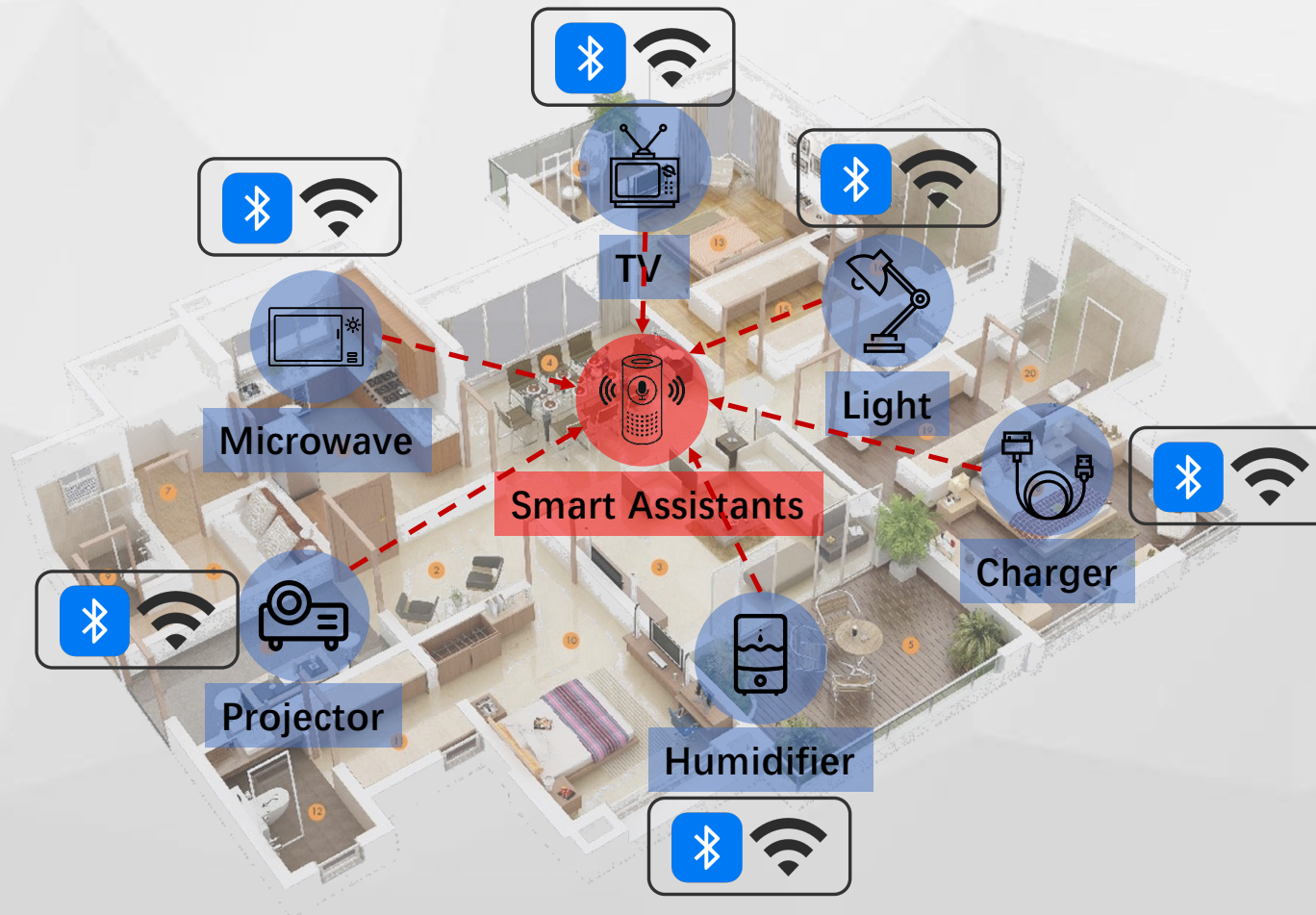


...

Motivation



Motivation



Motivation

Smart Assistants



What about those appliances **without** smart modules?



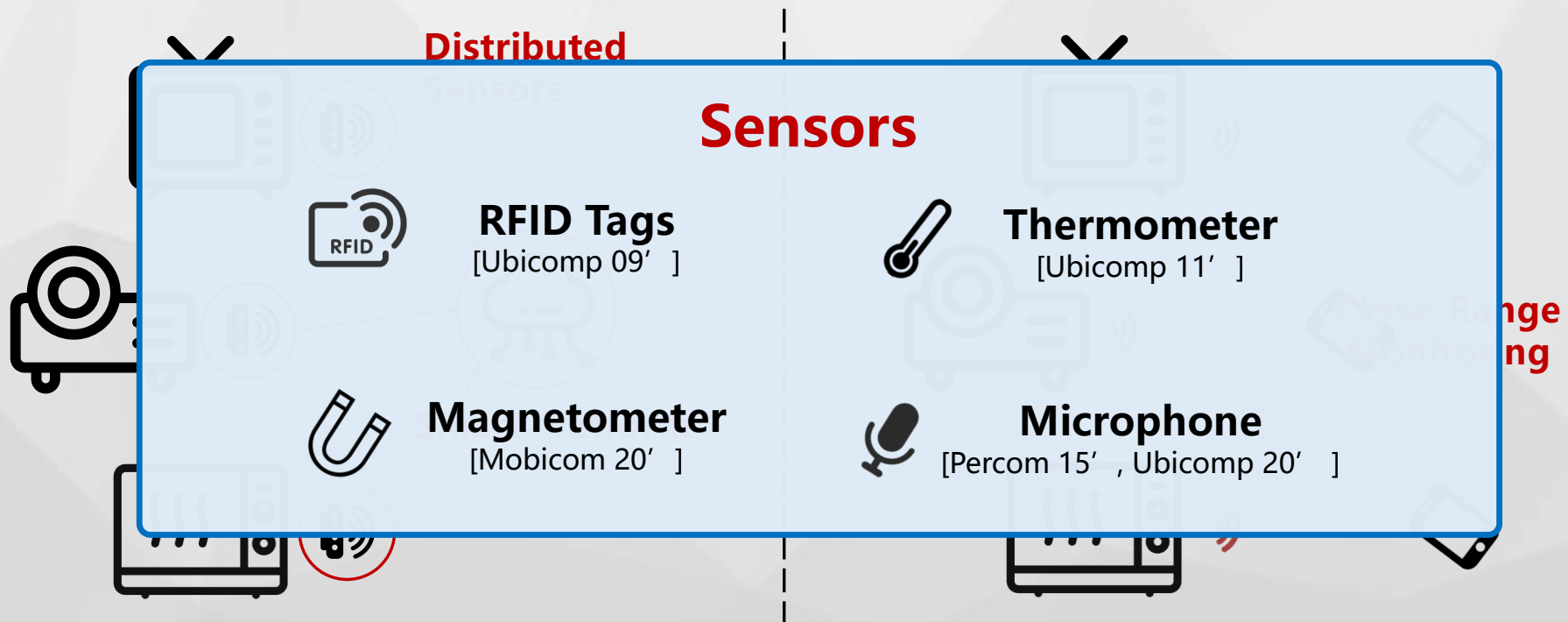
Smartphones



Smart Speakers

Existing Works

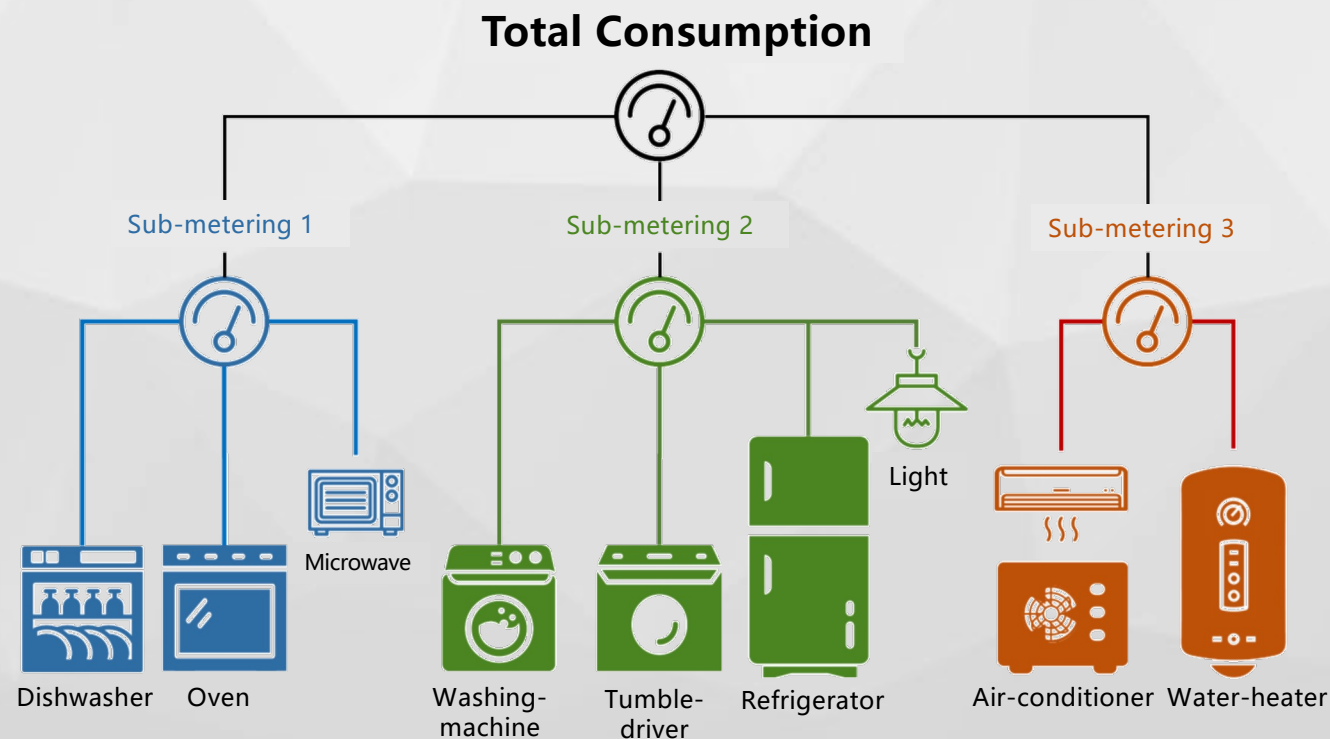
- **Close-Range Sensing:** install distributed sensors on each appliance or monitor appliances directly at a close range.



Additional Installation and Limited Range

Existing Works

- **Non-intrusive load monitoring (NILM):** tracking appliances by monitoring energy consumption based on total energy.



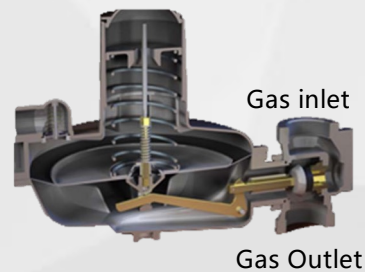
Failed for Time-varying and Same-model Devices

Existing Works

- **Infrastructure-mediated sensing (IMS)**: utilizes side-channels that are affected by the infrastructure of the house. (e.g., gas, water, vibration, and electricity.)

GasSense

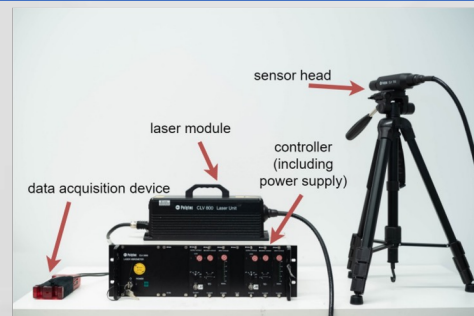
[Percom 10']



Gas

VibroSense

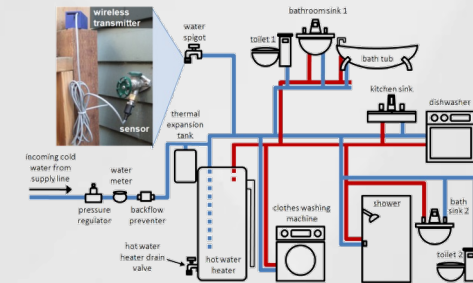
[Ubicomp 20']



Infrastructure Vibration

HydroSense

[Ubicomp 09']



Water

DOSE

[Percom 15']

ElectriSense

[Ubicomp 10']



Electricity

Additional Expensive or Customized Hardware

Limited Range

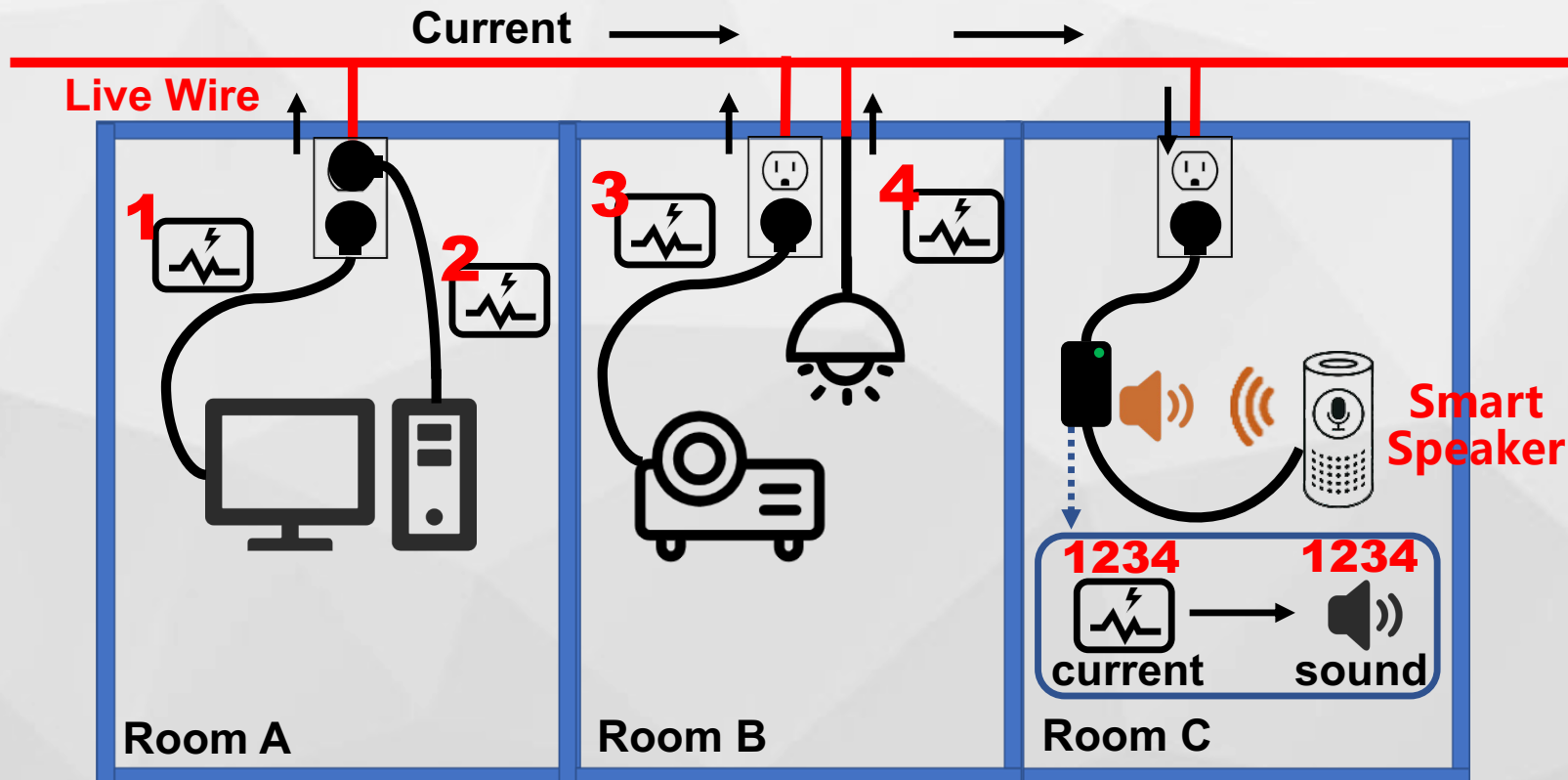
Limited Device Types

Additional Hardware

How to break these limitations?

Leveraging **Current-to-Acoustic** Channel

Our Proposed System - **AudioSense**



**Remote
Detection**

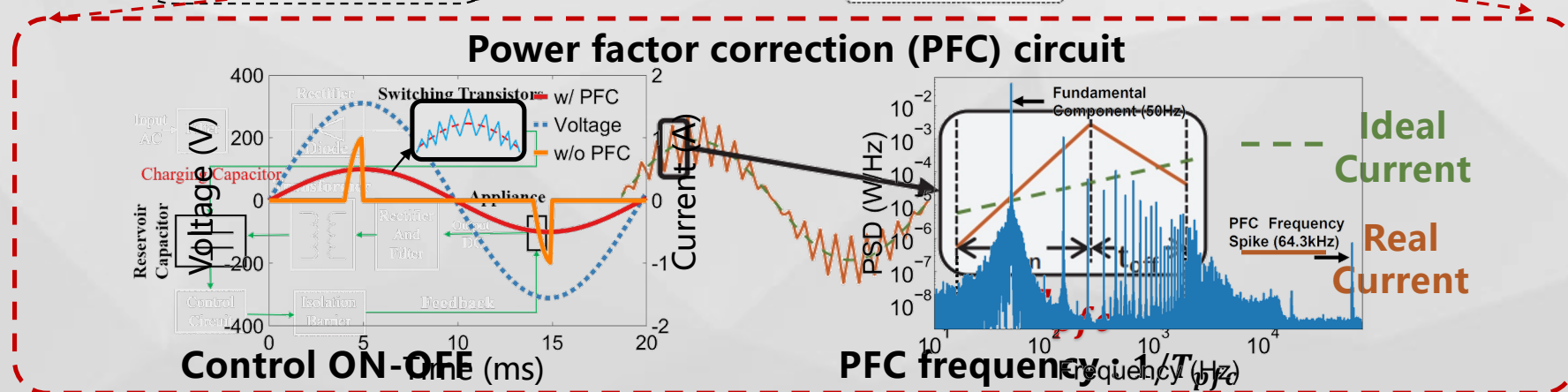
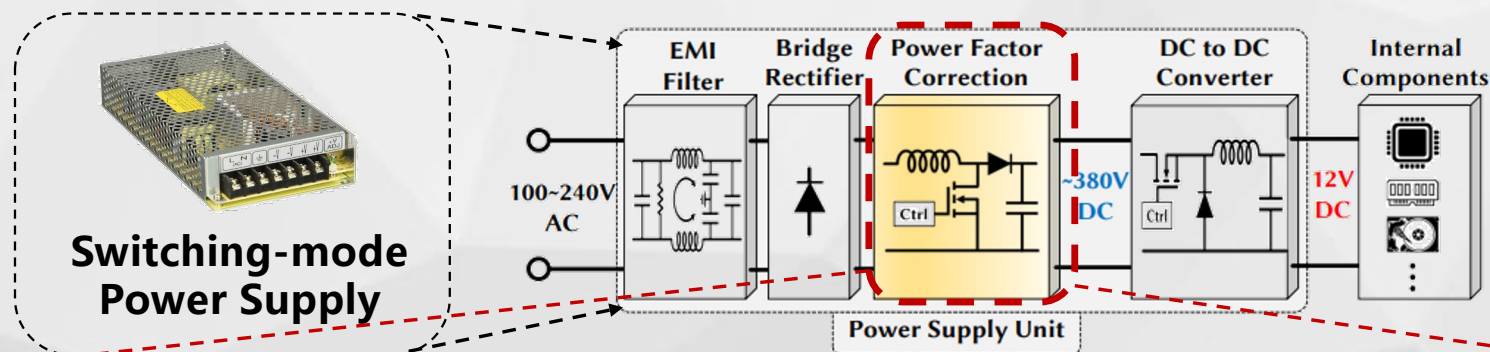
Single-Point

**Multiple
Devices**

**No Additional
Hardware**

Working Principle 1 – Power Factor Correction Circuit

Q1: Which feature to be utilized to identify electrical appliances ?



The appliance generates a current of specific **PFC switching frequency**

Working Principle 1 – Power Factor Correction Circuit



Electrical Devices

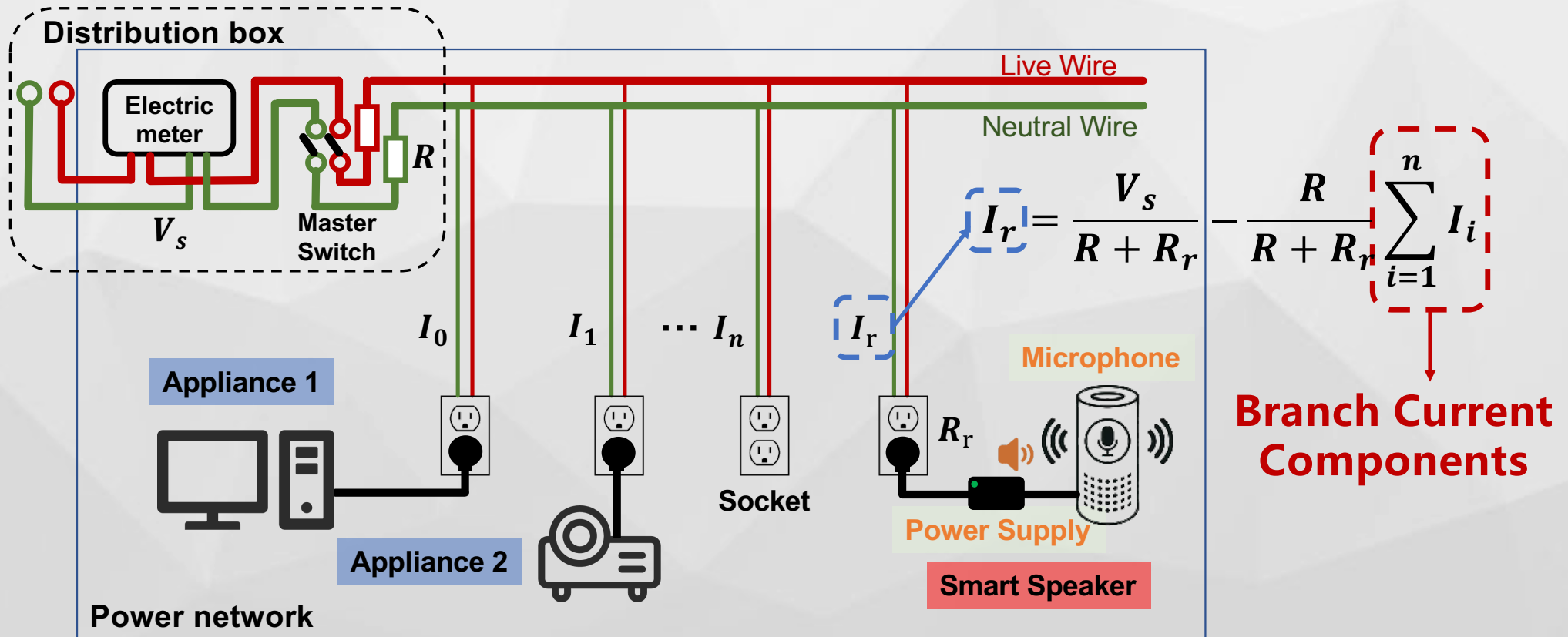
**According to IEC61000-3-2 [1],
PFC circuits must be installed in:**

- ◆ Lighting equipment ($> 5W$)
- ◆ Class D equipment ($> 300W$)
(e.g., monitors, TVs, computers)
- ◆ Chargers

The appliances containing PFC modules are **common in daily life**

Working Principle 2 – Powerline Interference

Q2: Why the feature can be detected remotely?

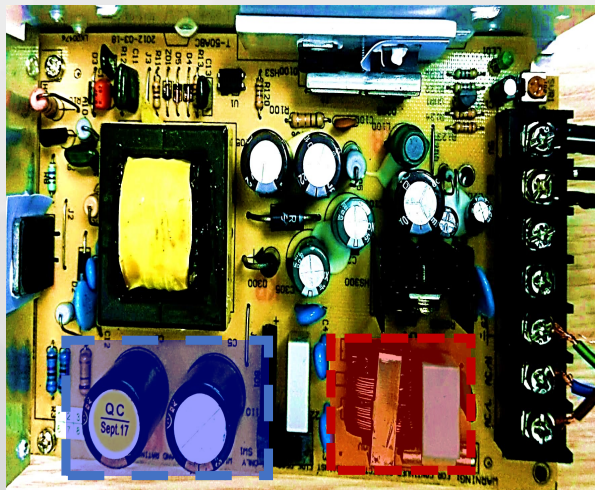


Within the same distribution box,
the current of each appliance interferes the branch of the smart speaker

Working Principle 3 – Acoustic Emission

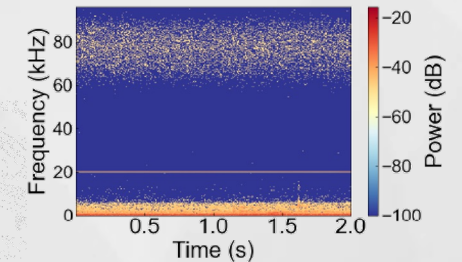
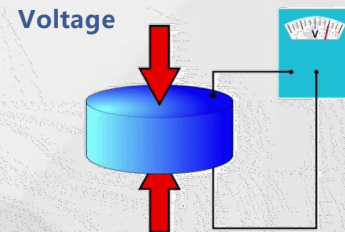
Q3: Why the power supply can emit sound? **Current-to-Acoustic Channel !**

Power Supply



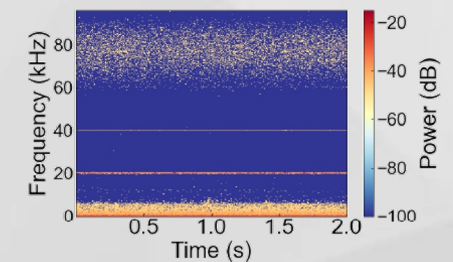
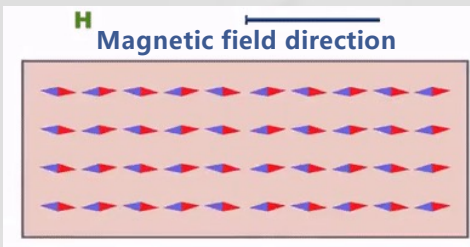
Capacitor Inductor

Capacitor



Deformation/Vibration due to **inverse piezoelectric effect**

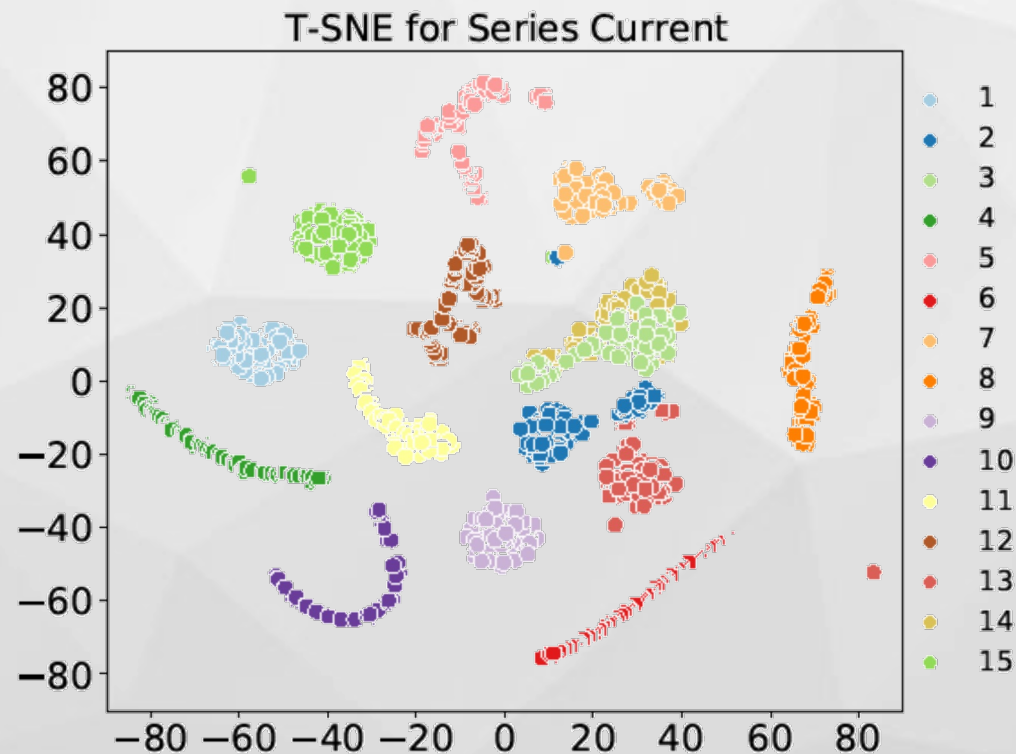
Inductor



Deformation/Vibration due to **magnetostrictive effect**

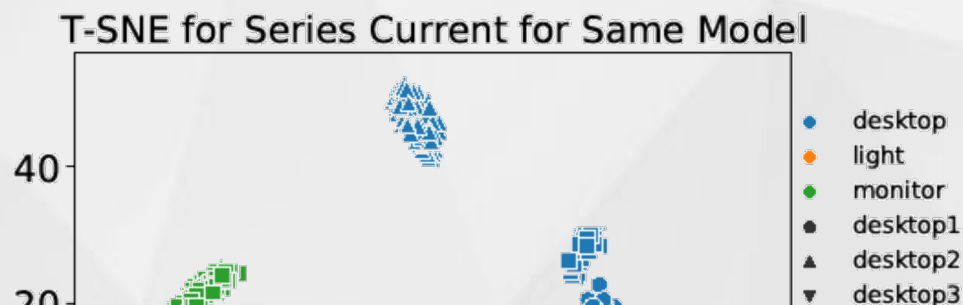
The power supply generates **sounds** as the **same frequency** of the current

Preliminaries - Uniqueness

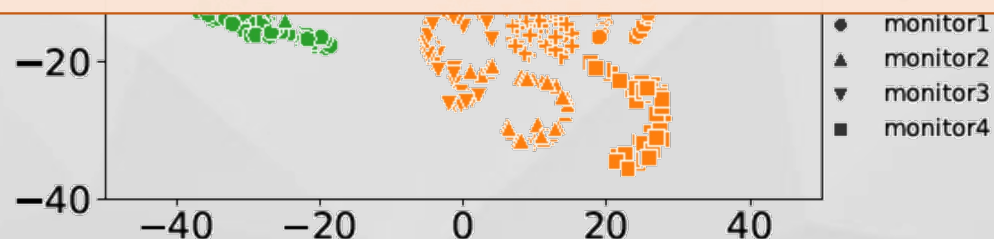


T-SNE of PFC signals among 15 different appliance models

Preliminaries - Uniqueness

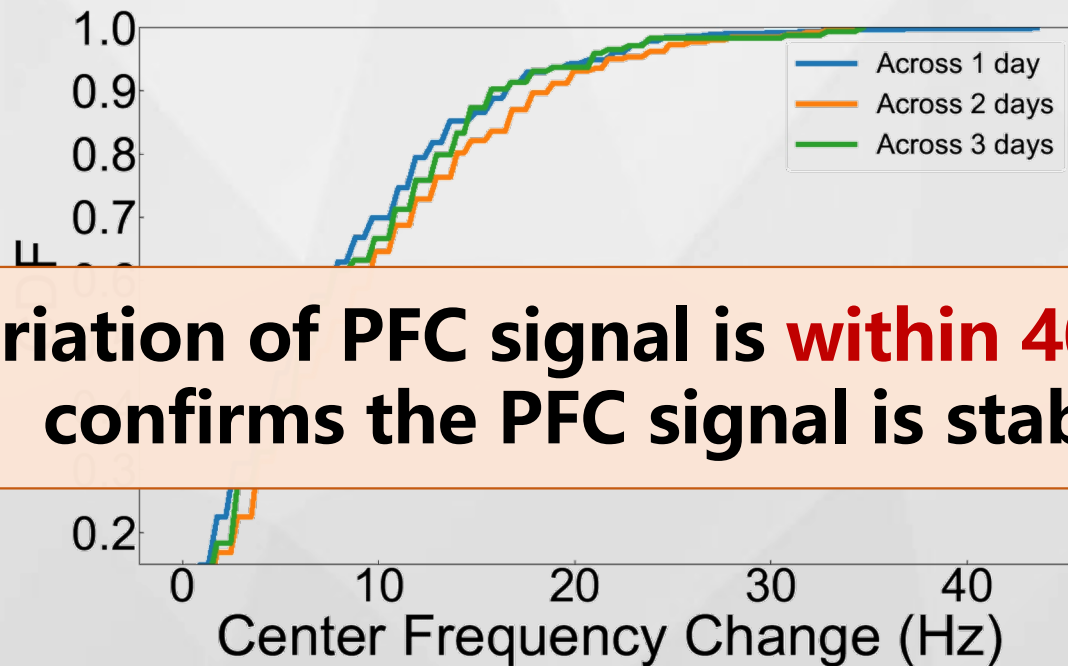


PFC frequencies of both **different** and the **same** model of appliances are **distinguishable**



T-SNE of PFC signals among the same model
(including 5 lights, 4 desktops, and 4 monitors)

Preliminaries - Stability

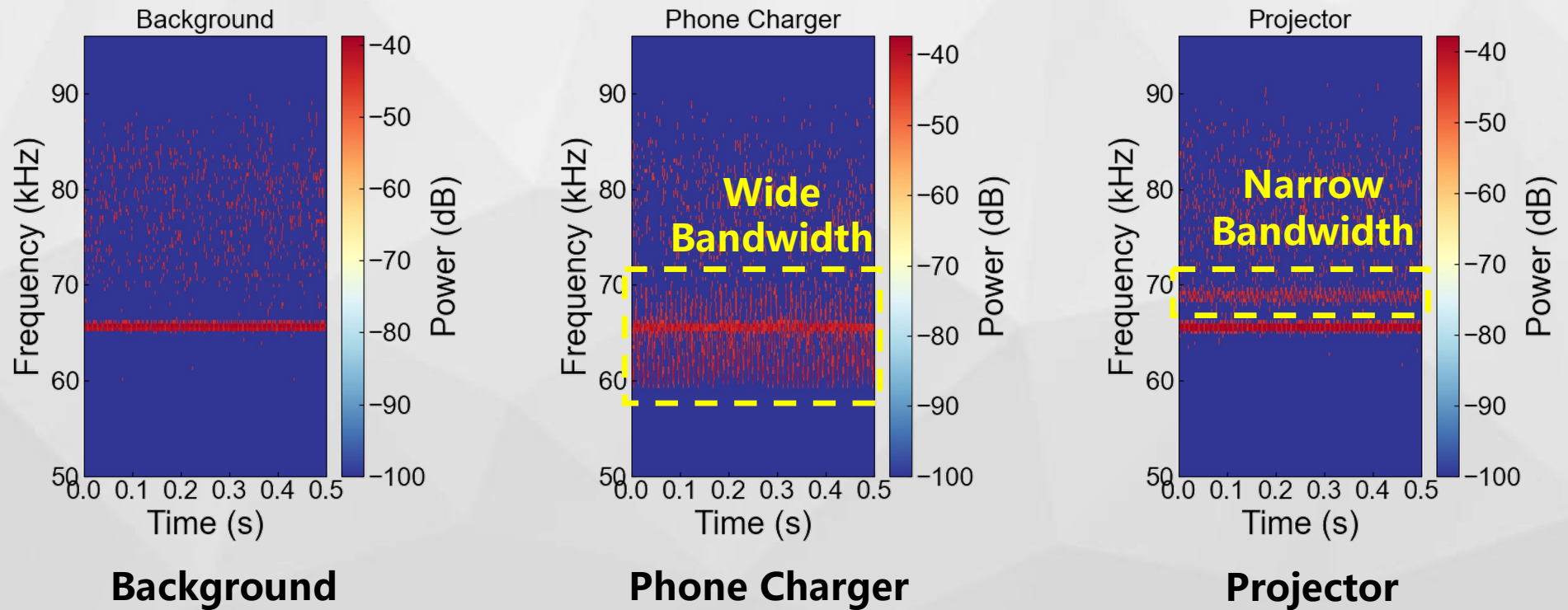


The variation of PFC signal is **within 40Hz** which confirms the PFC signal is stable

The CDF of PFC center frequency over various durations

Pre-Analysis of Audio Signal

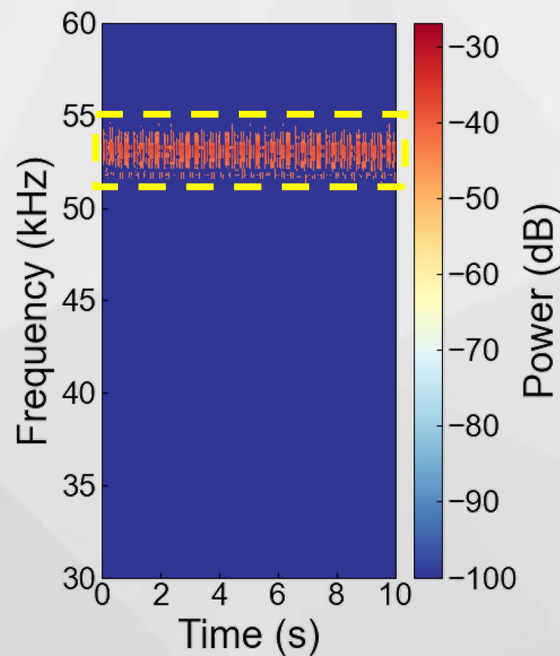
- Impact of type of devices



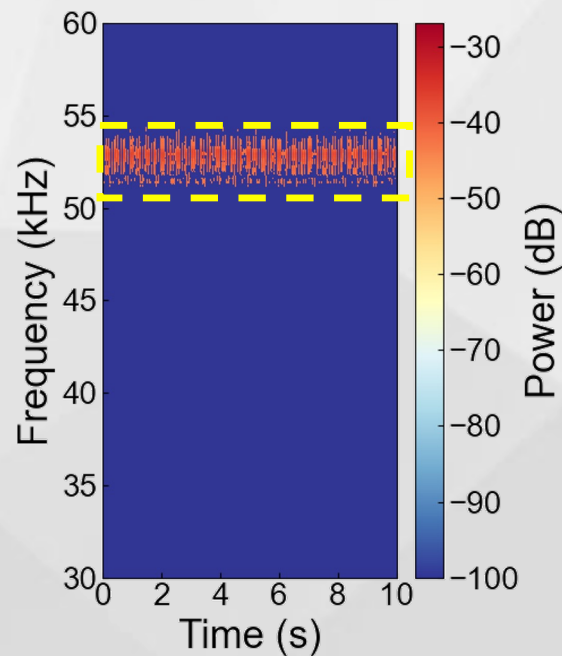
PFC signals present **different bandwidths** in different appliances

Pre-Analysis of Audio Signal

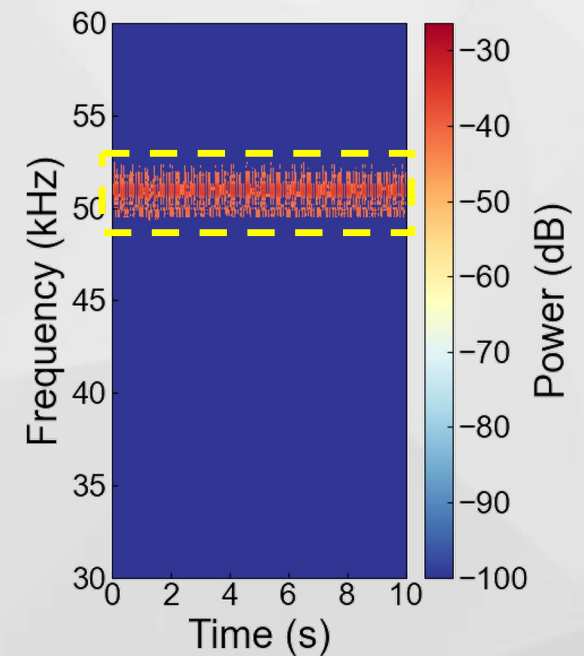
- Impact of working states: different power levels of an induction cooker



Low Power



Medium Power

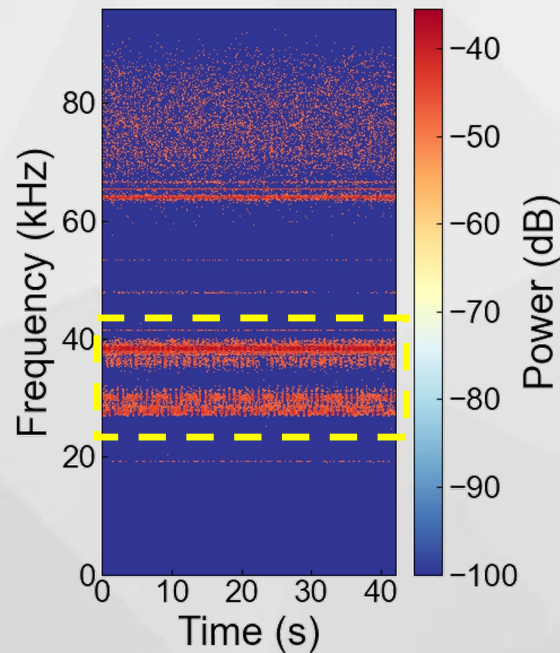


High Power

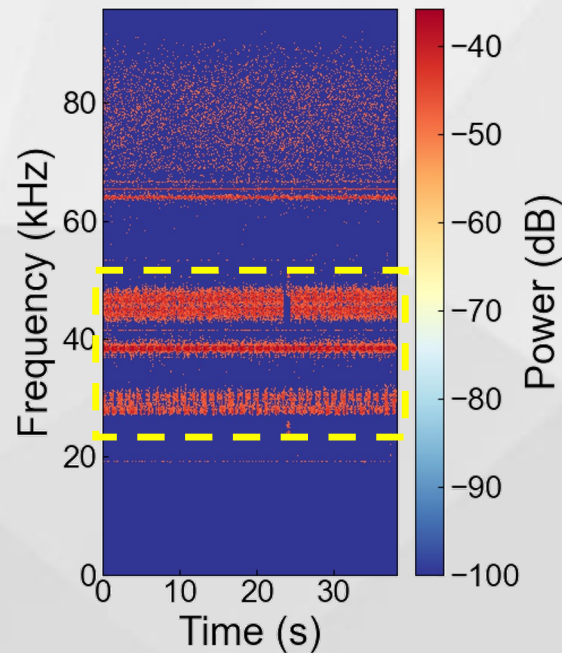
PFC signals present **different frequencies** in different working states

Pre-Analysis of Audio Signal

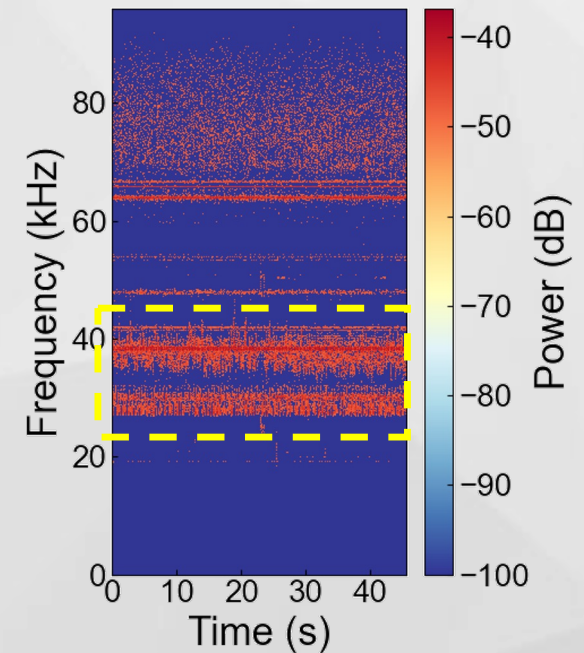
- Impact of charging states: different charging states of a smart phone



Battery <20%



Battery 20%-80%



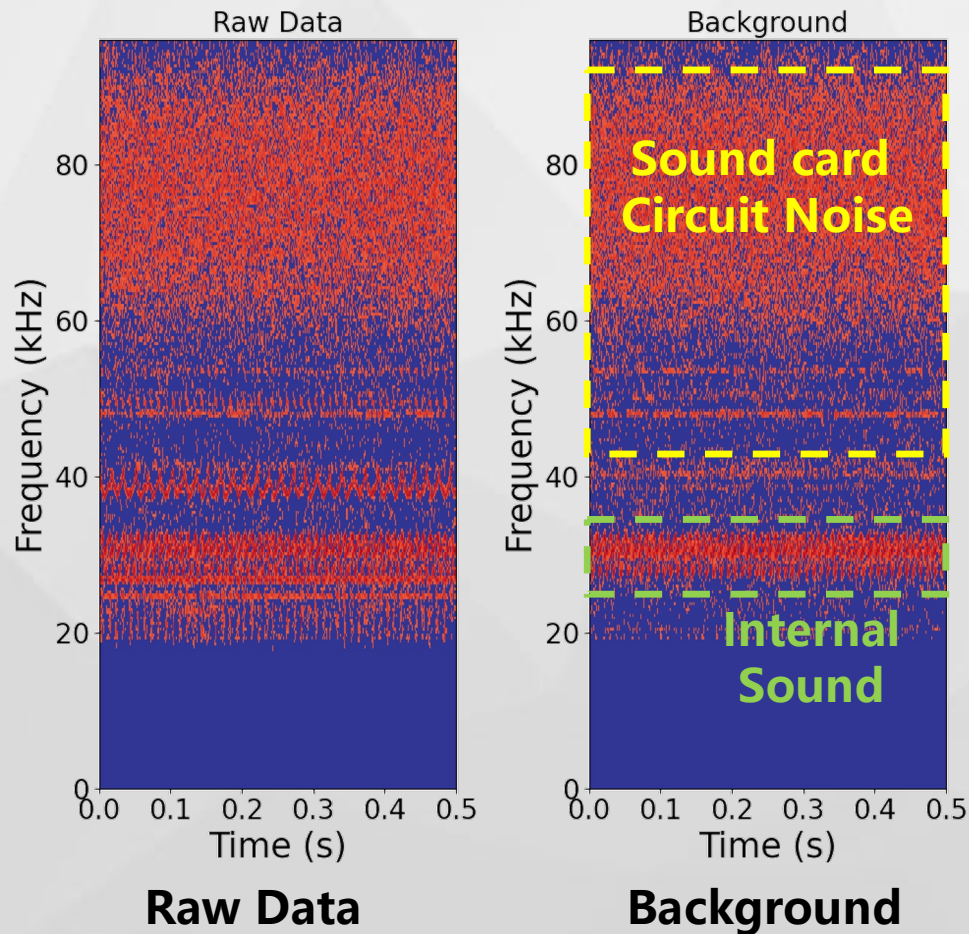
Battery >80%

PFC signals present different characteristics in different charging states

PFC signals are **periodic** signals of different **frequencies** and **bandwidths**

How to extract it effectively ?

Methodology – Spectral Subtraction



$$Y_r = \underbrace{S_{pfc}(\omega_0)}_{\text{PFC Components}} + \underbrace{N_1(\omega_1)}_{\text{Sound card Circuit Noise}} + \underbrace{N_2(\omega_2)}_{\text{Internal Sound}}$$

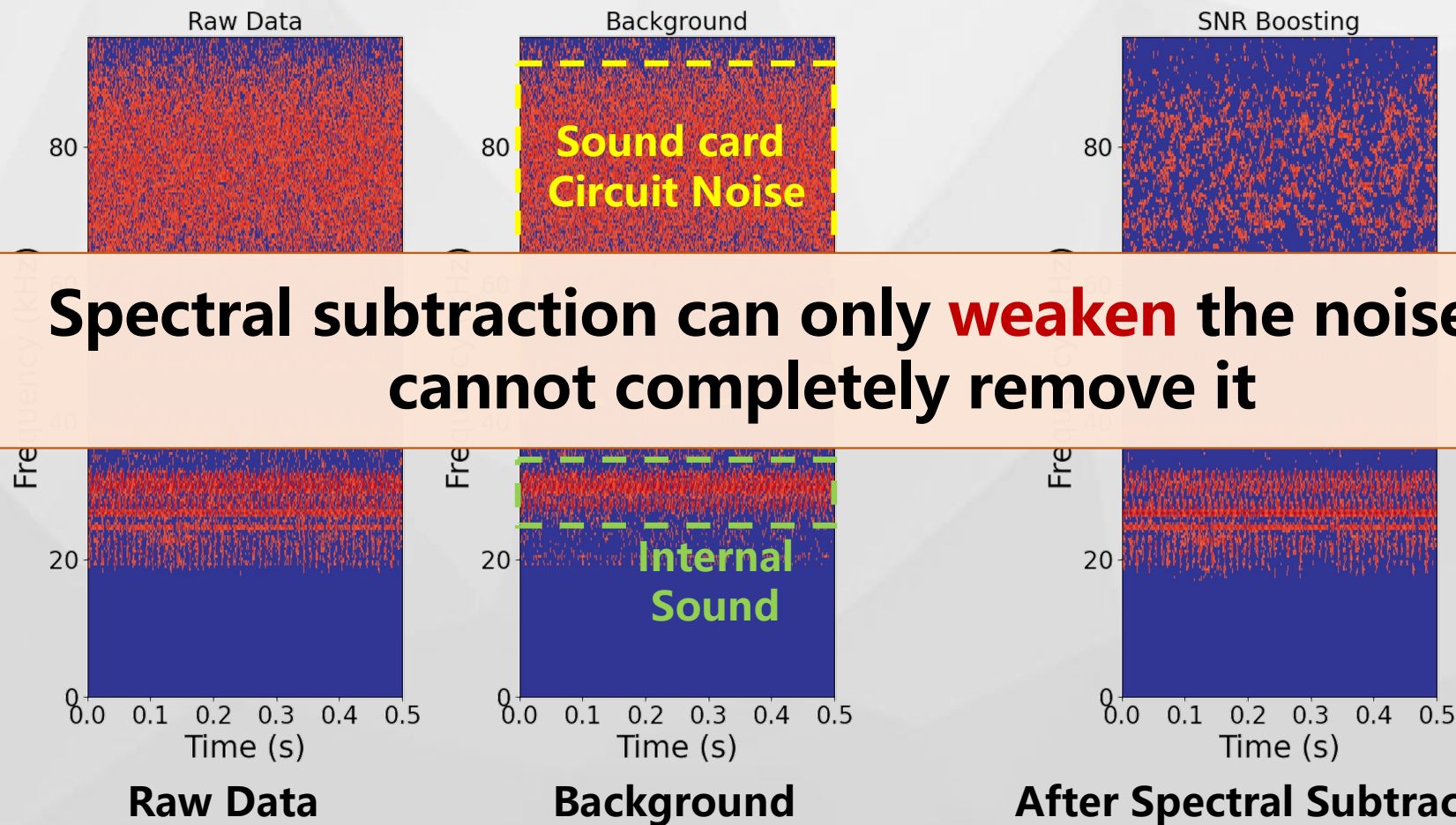
Y_{bg}

Spectral Subtraction:

$$\begin{aligned}\hat{S}_{pfc}(\omega_0) &= F^{-1} \left[\left[|Y_r| - E[|Y_{bg}|] \right] e^{j\theta} \right] \\ &= S_{pfc}(\omega_0) + \epsilon\end{aligned}$$

E is the estimation of Y_{bg} ,
 F^{-1} is inverse fast Fourier transform
 ϵ is the estimation error.

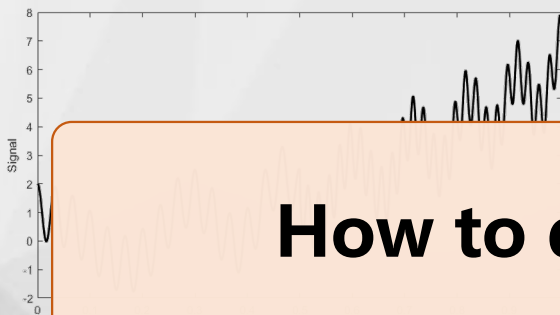
Methodology – Spectral Subtraction



Methodology - Optimized VMD

- Traditional variational mode decomposition (VMD) [1]

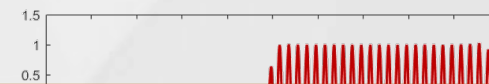
Original Signal



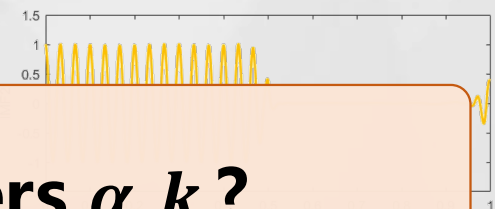
α, k

Intrinsic Mode Functions (IMFs)

IMF 1



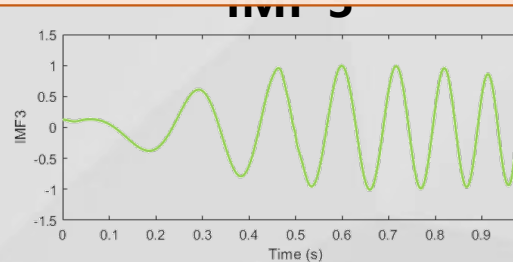
IMF 2



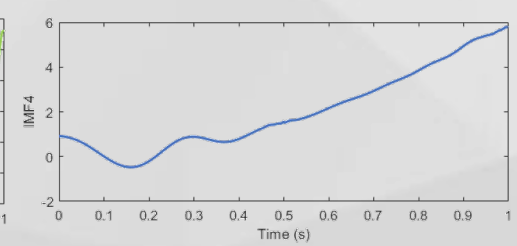
How to determine the best parameters α, k ?

$$\min_{u_k(\omega_0)} \left\{ \left\| \sum_k u_k(\omega_0) - \hat{S}_{pfc}(\omega_0) \right\|_2^2 + \alpha \sum_k \|j\omega u_k(\omega_0 - \omega_k)\|_2^2 \right\}$$

k is the number of IMFs
 α controls the bandwidth of IMFs
 ω is the center frequency.



IMF 4



Smaller α leads to larger bandwidth, and larger α leads to mode splitting
Smaller k leads to mode loss, and larger k leads to mode mixing.

Methodology - Optimized VMD

- **Bandwidth-based optimization for VMD**

Key idea: inappropriate k or α leads to a **larger** sum of bandwidth **BW**

$$\Delta_f^2 = \int_0^\infty (\omega - \omega_k)^2 \frac{|\hat{u}_k(\omega)|^2}{\int_0^\infty |\hat{u}_k(\omega)|^2 d\omega} d\omega$$
$$\text{BW} = \sum_{k=1}^k bw_k = \sum_{k=1}^k 2\Delta_f$$

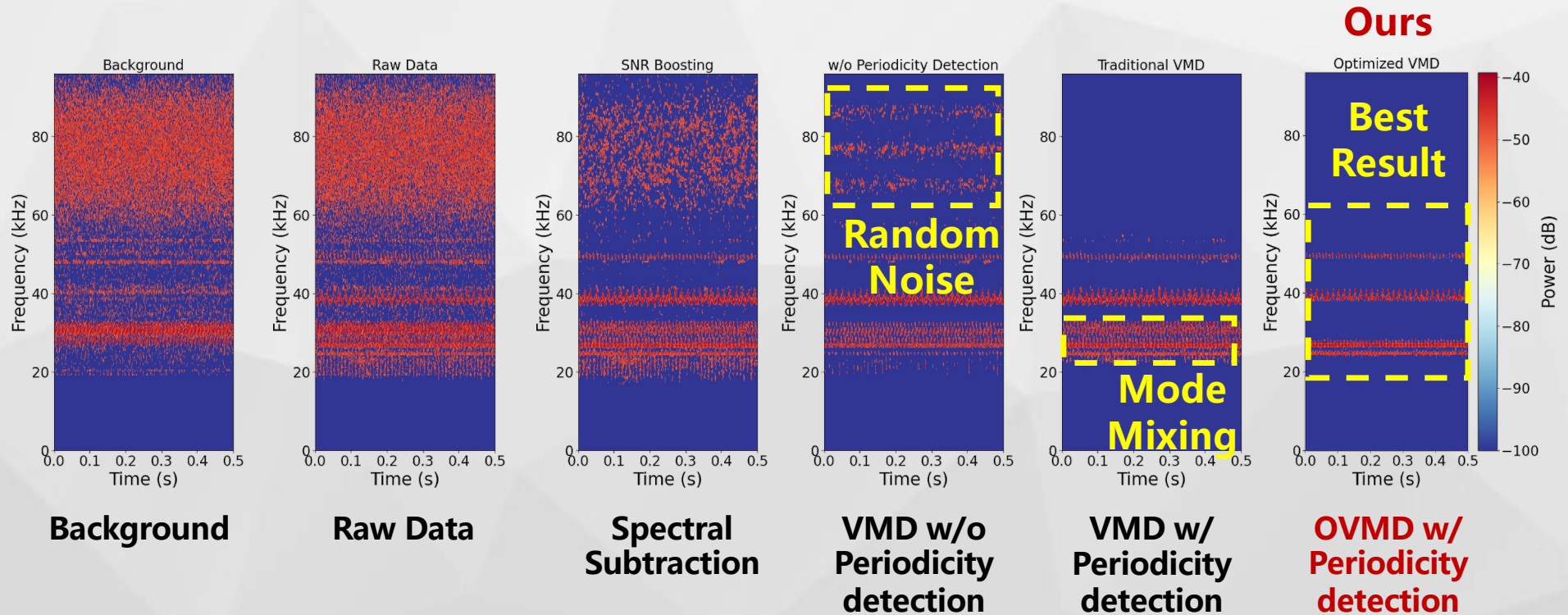
We choose the best combination of $k \in [2,40]$ and $\alpha \in [1e3,1e5]$ to achieve the **minimal** BW

- **Periodicity Detection**

Key idea: PFC signals are **periodic** signals while the noise floor of the sound card is **non-periodic** (i.e., random noise)

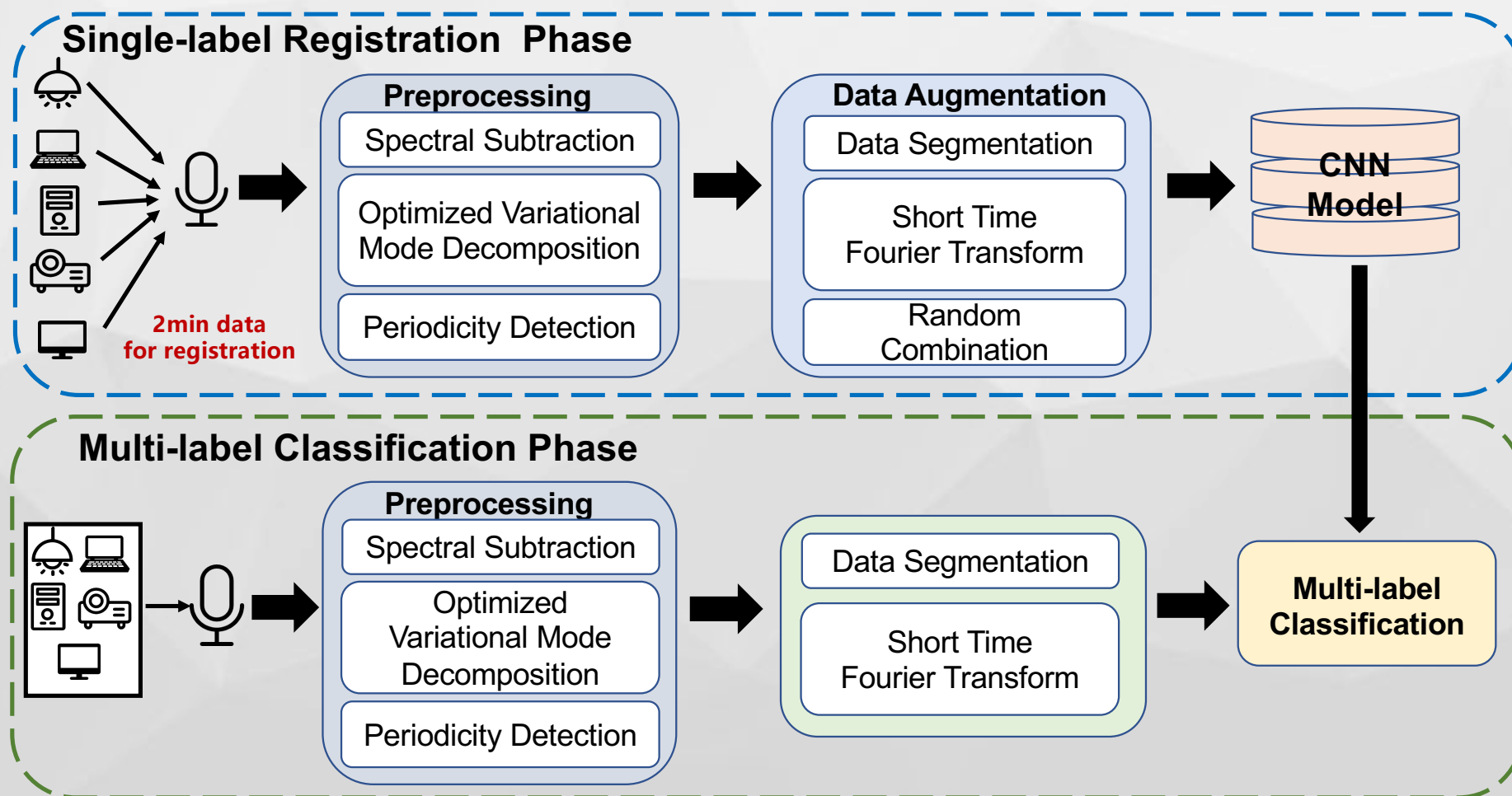
We calculate the **upper envelope** and derive the **auto-correlation coefficients** of each IMF to find the IMF that is a periodic signal

Methodology

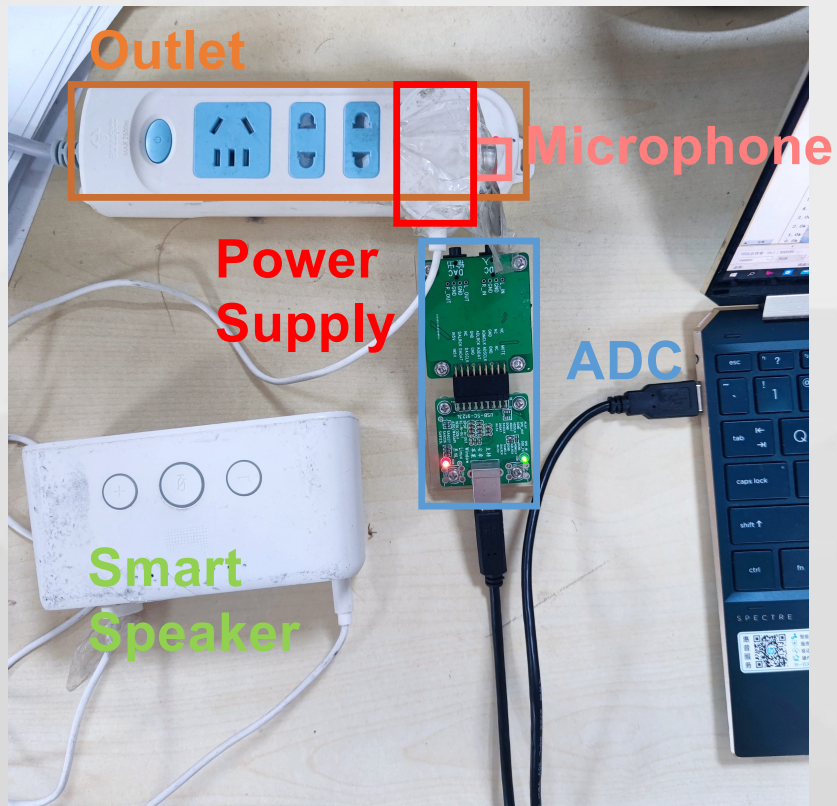


Our proposed **optimized VMD** can cleanly extract the desired PFC signal

System Architecture



Experiment Setup

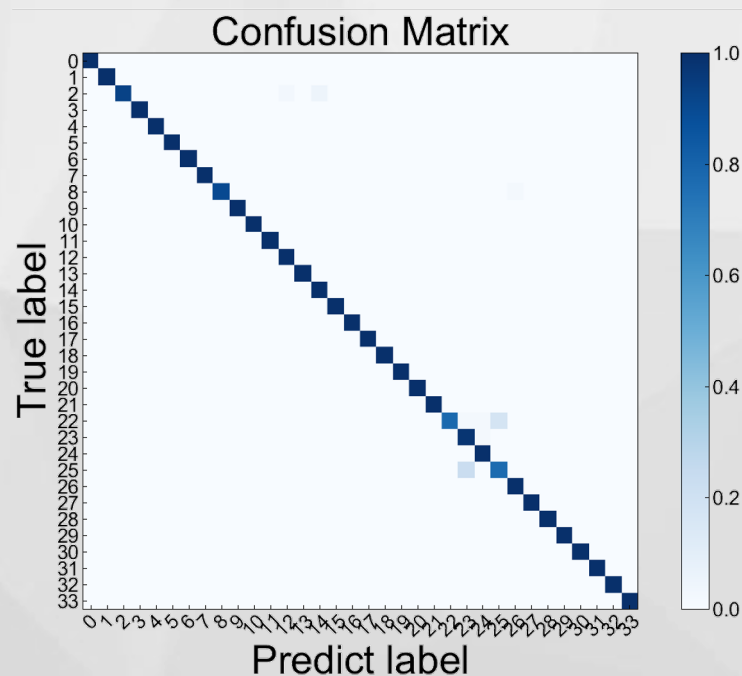


Appliance	Company	#States	#Same	#Different	#Total num	#Total states
Light	Xiaomi, PHILIPS, etc.	1	5	2	7	7
Monitor	Dell, PHILIPS, etc.	1	4	1	5	5
Desktop	Dell, HP, etc.	1	4	2	6	6
Laptop	Hp, Lenovo	1	0	2	2	2
Projector	Canon, Sony	1	0	2	2	2
Phone charger	Huawei, Xiaomi	3	0	2	6	6
Induction cooker	Supor	3	0	1	1	3
TV	Skyworth	1	0	1	1	1
UPS	SANTAK	1	0	1	1	1
Accumulator	Chilwee	1	0	1	1	1

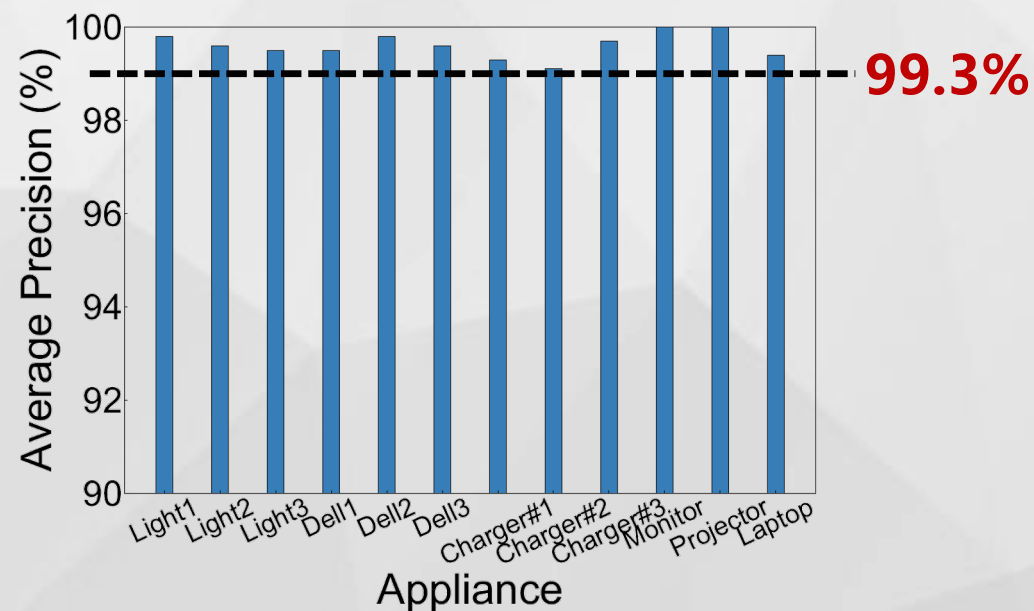
28 Appliances with 34 states

Performance Evaluation

- Overall Performance



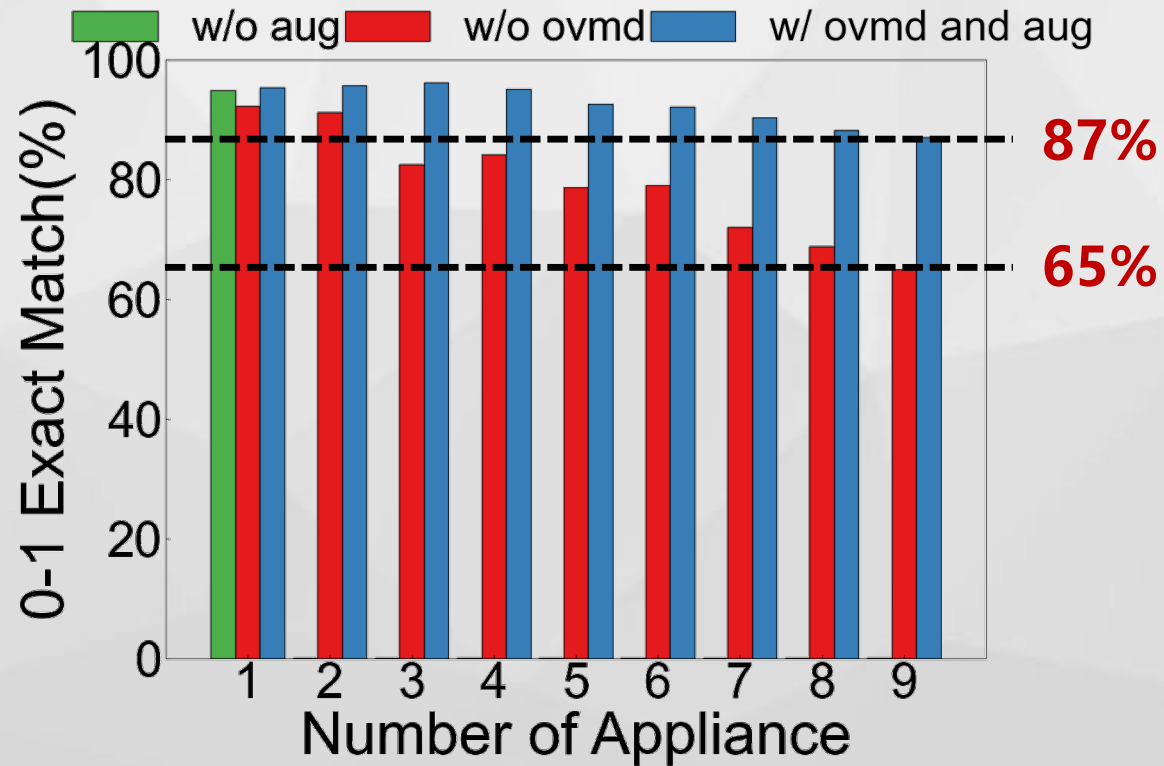
Single-label Classification



Multi-label Classification

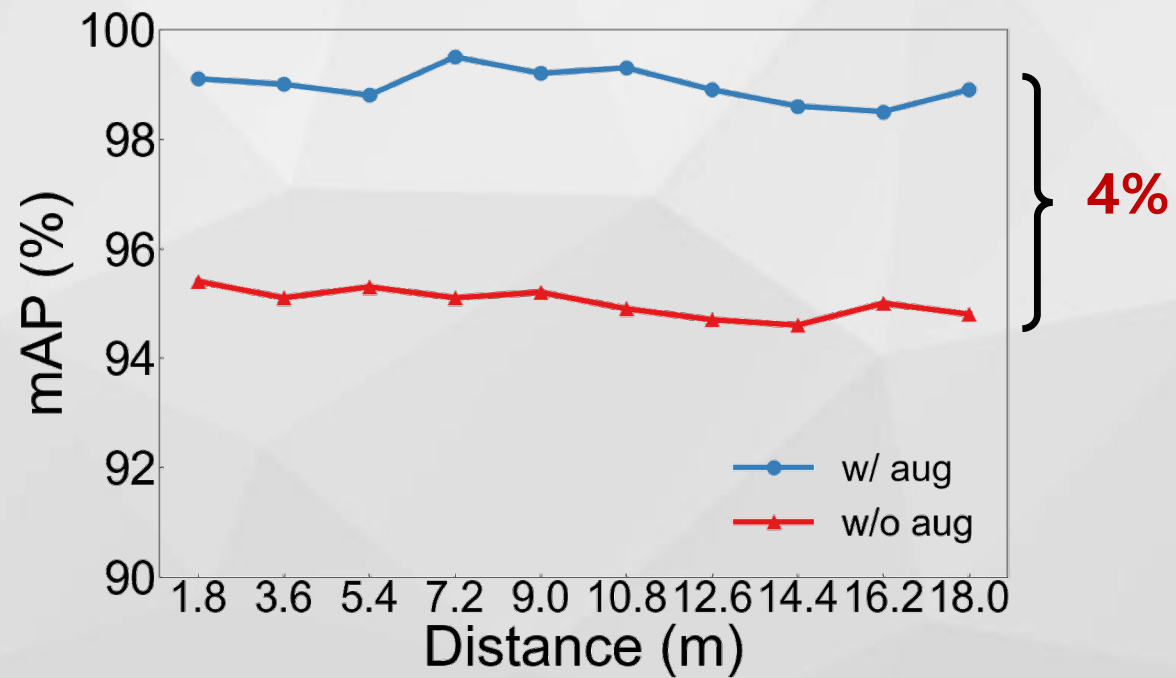
Performance Evaluation

- Impact of number of appliance



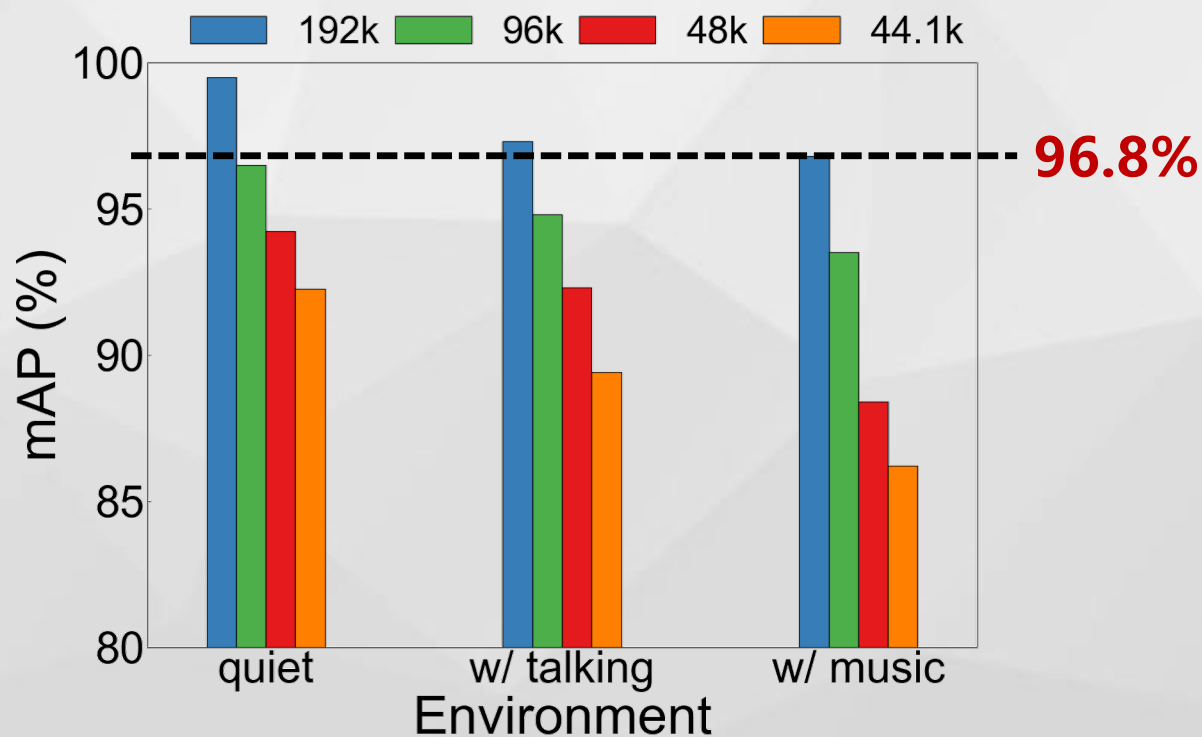
Performance Evaluation

- Impact of distance attenuation



Performance Evaluation

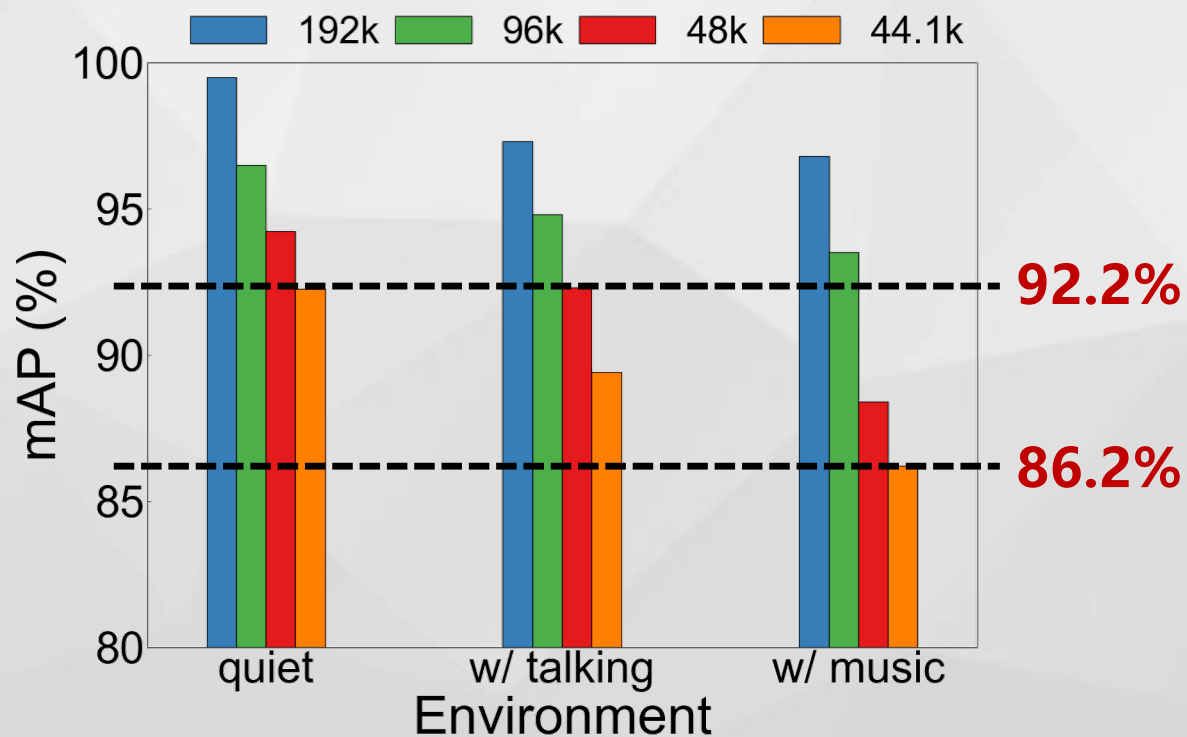
- Impact of environment interference



All mAP values exceeded **96.8%** when the sampling rates is 192kHz

Performance Evaluation

- Impact of environment interference



For 44.1kHz, mAP achieves **92.2%** and **86.2%** in quiet and music environment, respectively

Demo

Conclusion

- We proposed a novel study to use **the sound** generated by power supplies to **remotely** identify electrical appliances **at a single point** based on a **current-to-acoustic model**.
- We developed an **Optimized Variational Mode Decomposition (OVMD)** scheme to adaptively optimize the extraction of PFC features.
- We developed a multi-label classification model from single-label data by using a **data augmentation scheme** to improve usability.
- We built the prototype of AudioSense. The overall performance achieved the mean average precision of **99.3%**.

Thanks For Watching!

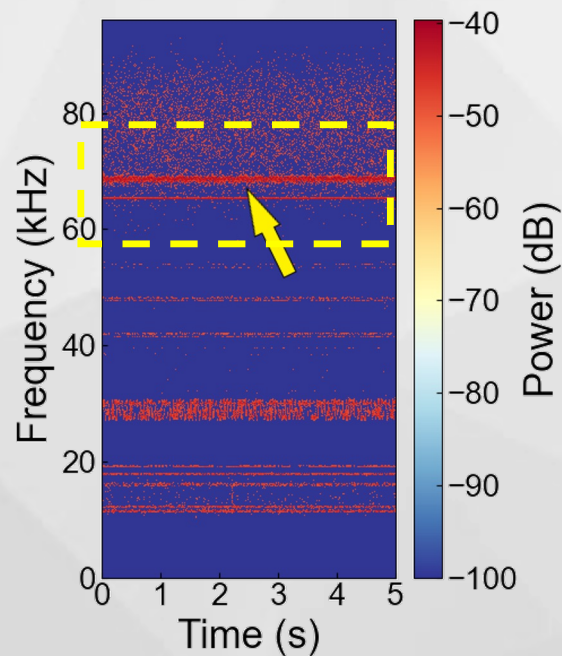
Yijie Li: yijieli@sjtu.edu.cn
Shanghai Jiao Tong University



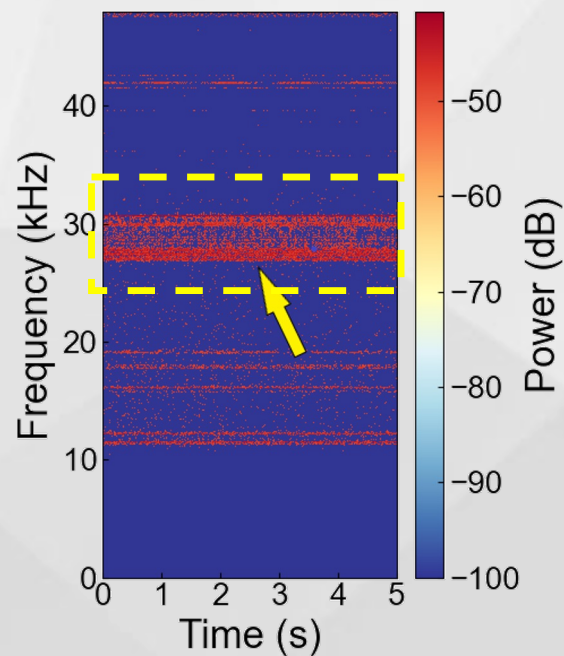
Backup Slides

Pre-Analysis of Audio Signal

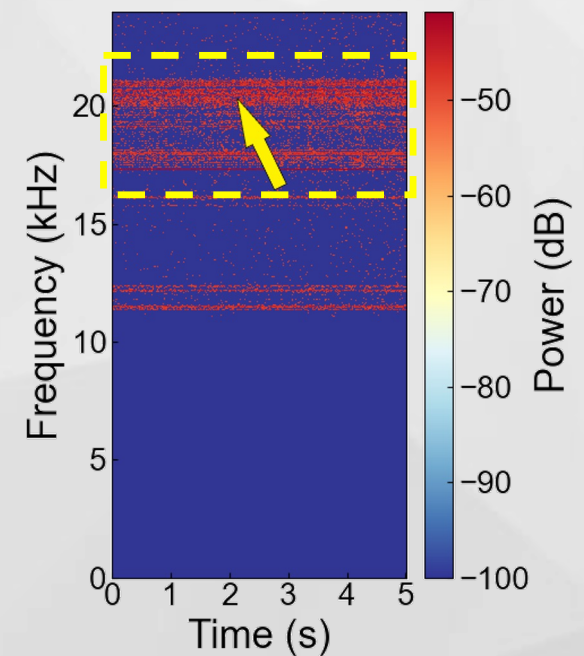
- Impact of sampling rates



192kHz



96kHz



48kHz

Aliasing occurs when sampling in insufficient sampling rates

Pre-Analysis of Audio Signal

- Impact of sampling rates

HIFI Device



Redmi K20pro premium (2019)
96kHz



LG V60 (2020)
192kHz

- Motorola Edge 2020 – **192kHz** microphone
- Samsung Galaxy S9+ - **192kHz** microphone
- Xiaomi Redmi Note9 Pro – **192kHz** microphone
- Samsung Galaxy S10 – **192kHz** microphone

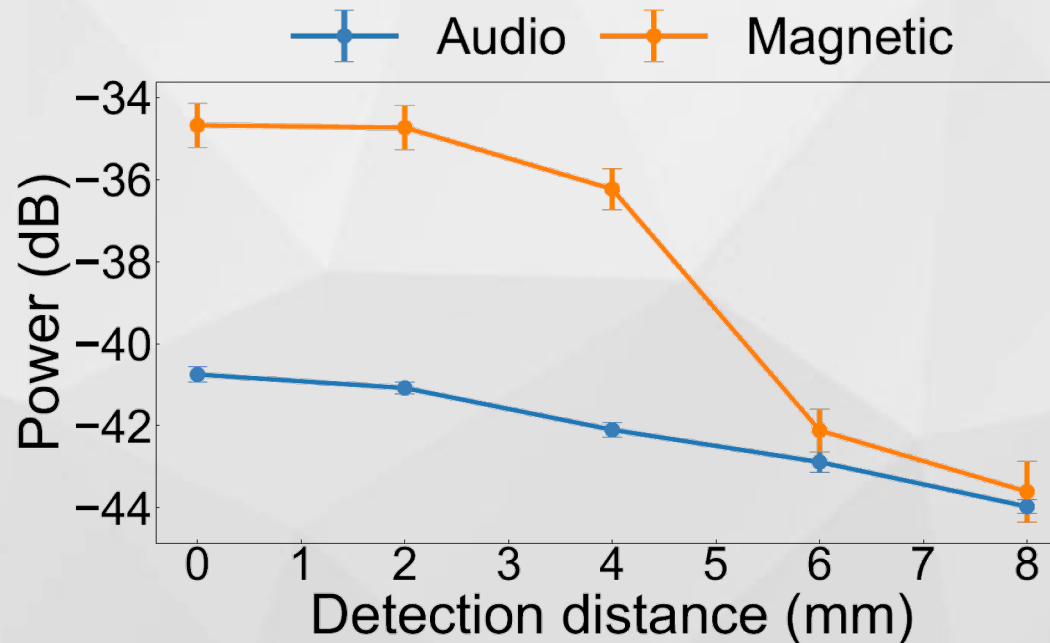
COTS smartphones[1]

Table 1: List of RX and TX capabilities for the off-the-shelf smartphones.

Brand	Lineup	# of Models	Audio Codec	RX f_s	Speaker Amplifier	TX f_s
Google	Pixel 4	1	Knowles IA8505	192 kHz	Cirrus Logic CS35L36	384 kHz
Google	Pixel 4a/5/5a	3	Realtek RT5514	192 kHz	Cirrus Logic CS35L41	192 kHz
Samsung	Galaxy S21 lineup (Exynos)	4	Exynos 2100 (integrated)	384 kHz	Cirrus Logic CS35L41	192 kHz
Samsung	Galaxy A50 lineup	2	Exynos 9610 (integrated)	384 kHz	Cirrus Logic CS35L41	192 kHz
Samsung	Galaxy A10/A20/A30/A40 lineup	10	Samsung COD3035x	192 kHz	Silicon Mitus SMA1301	192 kHz
Samsung	Galaxy S9 lineup (Snapdragon)	2	Qualcomm WCD9341	192 kHz	Maxim MAX98512	192 kHz
Samsung	Galaxy S10 lineup (Exynos)	5	Cirrus Logic CS47L93	192 kHz	Cirrus Logic CS35L40	192 kHz
Xiaomi	Redmi K40 lineup	4	Qualcomm WCD9380	192 kHz	Cirrus Logic CS35L41	192 kHz
Xiaomi	Mi 9/10/11/12/13 lineup	24	Qualcomm WCD93xx	192 kHz	Cirrus Logic CS35L41	192 kHz
Xiaomi	Redmi Note 8/8T/9Pro lineup	3	Qualcomm WCD937x	192 kHz	TI TAS2562	96 kHz
Huawei	P40 lineup	3	HiSilicon Hi6405	384 kHz	TI TAS2564	192 kHz
Huawei	Mate20/30/40, P30, Nova6/7 lineup	24	HiSilicon Hi6xxx series	192 kHz	NXP TFA9874	96 kHz
ASUS	Zenfone 9	1	Qualcomm WCD9380	192 kHz	Qualcomm WSA8835	384 kHz
Motorola	Edge (2020)	1	Cirrus Logic CS47L35	192 kHz	Cirrus Logic CS35L41	192 kHz
Sony	Xperia 1/5 II/III, Xperia 10 lineup	6	Qualcomm WCD93xx	192 kHz	Cirrus Logic CS35L41	192 kHz
Vivo	iQOO Neo3 5G	1	Qualcomm WCD9341	192 kHz	Maxim MAX98928	48 kHz
OnePlus	OnePlus 6 lineup	2	Qualcomm WCD9341	192 kHz	Maxim MAX98928	48 kHz
OnePlus	OnePlus 9 lineup	3	Qualcomm WCD9385	192 kHz	NXP TFA98xx series	48 kHz
Oppo	Realme X50 lineup	4	Qualcomm WCD9385	192 kHz	NXP TFA98xx series	48 kHz

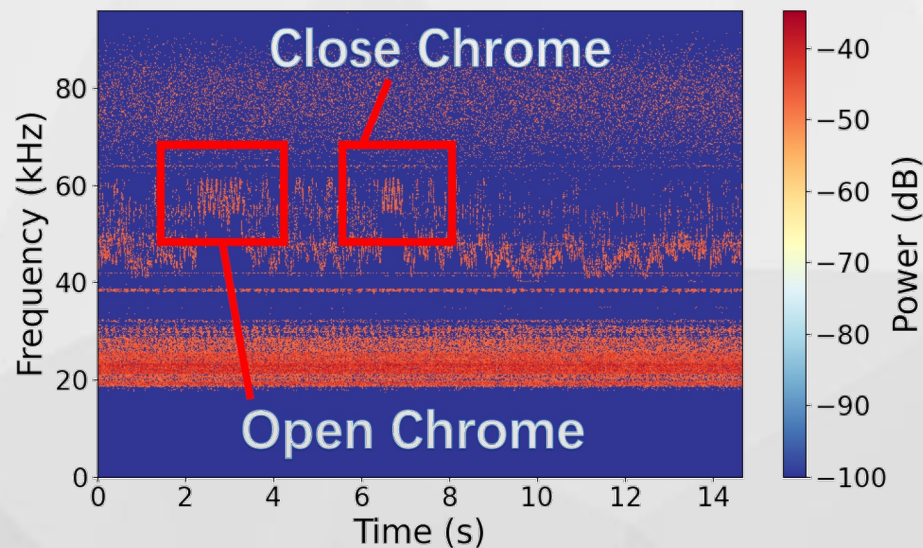
[1] PowerPhone: Unleashing the Acoustic Sensing Capability of Smartphones. Mobicom 2023

Comparison to Magnetic Signal



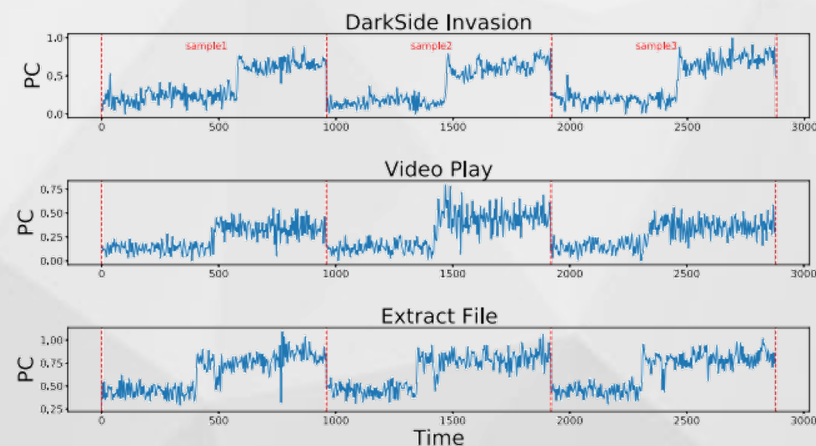
- Acoustic signals decays **linearly** and the **magnetic** signals decays cubically. **Acoustic signals can be detected at a greater distance**
- Most electronic equipment is equipped with **magnetic shielding**. (e.g., Faraday cage)
- Sampling rates. Ubiquitous magnetometer (50Hz) vs. Microphone (192kHz)

Extended Applications



Program Launching

- Detect EMI and classify the program launching process [1]
- Detect DarkSide Ransomware. [2]

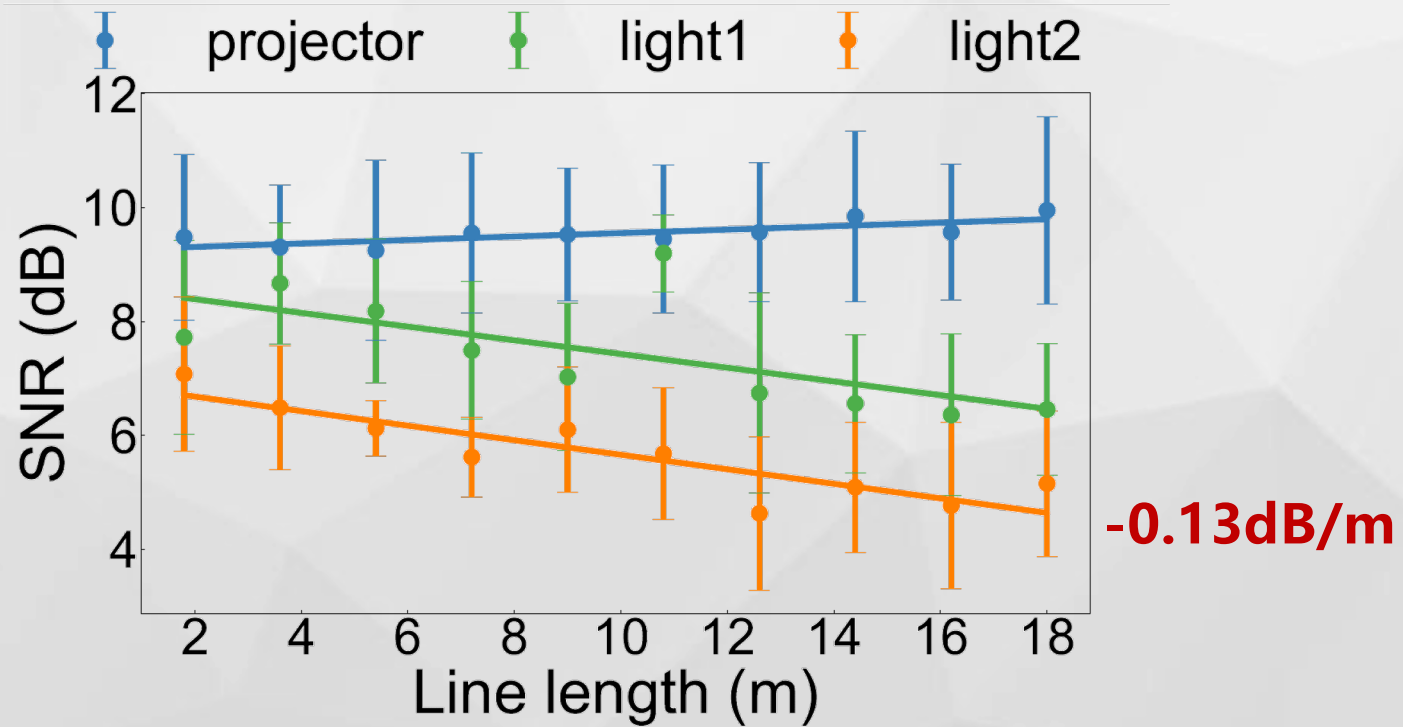


DarkSide Ransomware

[1] Zhang, Juchuan, et al. "Outletspy: cross-outlet application inference via power factor correction signal." *Proceedings of the 14th ACM Conference on Security and Privacy in Wireless and Mobile Networks*. 2021.

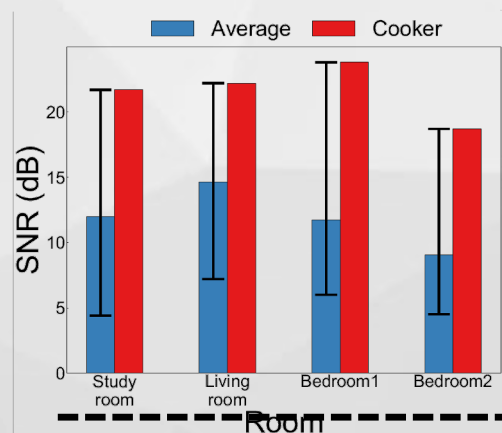
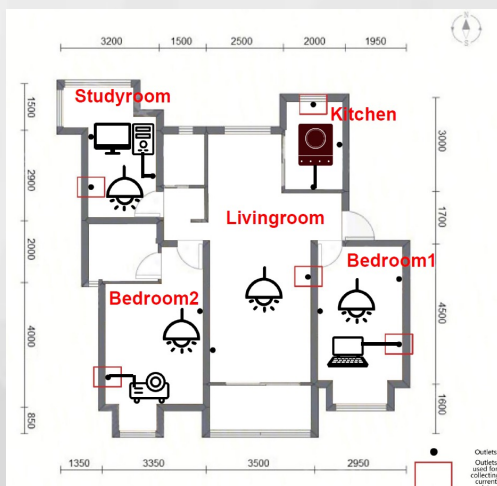
[2] Zou, Shan, et al. "OutletGuarder: Detecting DarkSide Ransomware by Power Factor Correction Signals in an Electrical Outlet." *2022 IEEE 28th International Conference on Parallel and Distributed Systems (ICPADS)*. IEEE, 2023.

Distance attenuation

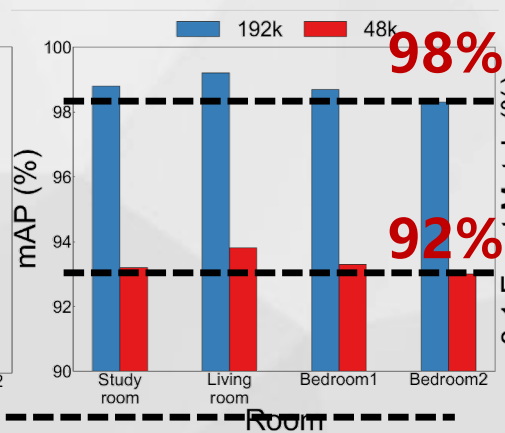


The maximum distance (line length) can achieve **53m**

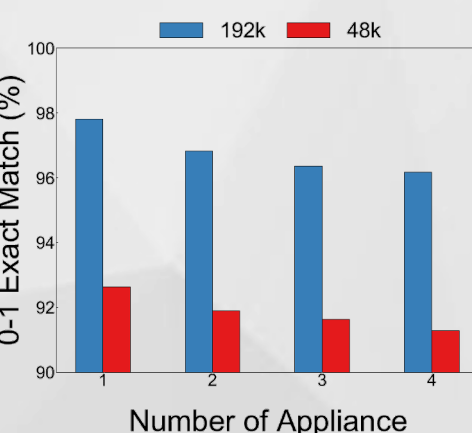
Case Study



**SNR in
different rooms**



**Overall
Performance**



**Number of
Appliance**

Quick Validation



Sound from a **desk lamp**'s power supply
is collected on a phone



Sound **Spectrogram**